

## No. of Functional Features Included in the Solution

|               |                                                        |
|---------------|--------------------------------------------------------|
| Date          | 15 March 2024                                          |
| Team ID       | XXXXXX                                                 |
| Project Name  | Credit Card Approval Prediction Using Machine Learning |
| Maximum Marks | 5 Marks                                                |

### Functional Features Overview:

| S.No | Feature Name                    | Feature Description                                                                                                | Module / Component   | Status | Marks Contribution |
|------|---------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------|--------|--------------------|
| 1    | User-Friendly Dashboard         | Provides an intuitive interface for navigating the application and accessing prediction features.                  | User Interface       | Done   | High               |
| 2    | Applicant Data Input            | Allows users to enter applicant details required for prediction.                                                   | Prediction Form      | Done   | High               |
| 3    | Data Validation                 | Validates user inputs before processing to ensure accuracy and completeness.                                       | Backend Validation   | Done   | Medium             |
| 4    | Credit Card Approval Prediction | Uses the trained machine learning model to predict whether a credit card application will be approved or rejected. | ML Prediction Engine | Done   | High               |
| 5    | Model Evaluation Dashboard      | Displays model performance using Accuracy, Confusion Matrix, and Classification Report.                            | Analytics Dashboard  | Done   | Medium             |
| 6    | Responsive Web Design           | Ensures the application works properly on desktops, tablets, and mobile devices.                                   | Frontend             | Done   | Medium             |
| 7    | Error Handling                  | Handles invalid inputs and unexpected system errors with user-friendly messages.                                   | Backend              | Done   | Medium             |

|   |                        |                                                                                    |                    |      |     |
|---|------------------------|------------------------------------------------------------------------------------|--------------------|------|-----|
| 8 | GitHub Version Control | Maintains project source code and enables collaboration using a GitHub repository. | Project Management | Done | Low |
|---|------------------------|------------------------------------------------------------------------------------|--------------------|------|-----|

**Feature Summary:**

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 5             |
| Additional / Nice-to-Have Features | 3             |
| Features Tested & Verified         | 8             |

**Feature Category Breakdown:**

| S.No | Category                  | Features in Category | Example Features                                                                                       |
|------|---------------------------|----------------------|--------------------------------------------------------------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | Dashboard, Applicant Data Input Form                                                                   |
| 2    | Backend / Logic           | 3                    | Data Validation, ML Prediction Engine, Error Handling                                                  |
| 3    | Database / Storage        | 1                    | CSV Dataset & Trained ML Model (.pkl)                                                                  |
| 4    | API / Integration         | 1                    | Flask Integration with Machine Learning Model                                                          |
| 5    | Security / Authentication | 1                    | Admin Login and Session Management ( <i>or Input Validation if authentication is not implemented</i> ) |